From Weyl Anomaly to Defect Energy and Entropy

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Based on [ZXH-Yuan-Zhou, arXiv: 2501.09498]

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Weyl anomaly of CFT

Weyl anomaly (central charge) governs the **entanglement structure** and **ground state energy** of CFTs

- **Casimir energy** (low-temperature asymptotics of partition function)

$$d = 2 \qquad\qquad E = -\frac{c}{12}$$

- **Rényi entropy** (generalization of the entanglement entropy)

$$d = 2 \qquad\qquad S_n^{\text{2d CFT}} = \frac{c}{6}\left(\frac{1}{n} + 1\right)\log\frac{l}{\epsilon}$$

Defects in QFT

- Defects are usually defined by non-local operators/B.C. in QFT

- Line defects, surface defects, boundaries…

  Wilson loops, Wilson surfaces, interface/boundary;

  Topological defects, …

- Nontrivial interaction with the bulk D.o.F.

Surface Defects in CFT

Analogous to 2d CFT, surface defect has three anomaly coefficients:

$$\langle t^{\mu}_{\mu} \rangle = -\frac{1}{24\pi}\left[\, b\, R^{\Sigma} + d_1 \Pi^{\mu}_{ab}\Pi^{ab}_{\mu} - d_2 W^{ab}_{ab}\right]$$

- Analogous relation in the presence of defects?
- Any non-perturbative way to study defects?

Recall the 2d CFT results:

$$E = -\frac{c}{12}$$

$$S_n^{\text{2d CFT}} = \frac{c}{6}\left(\frac{1}{n}+1\right)\log\frac{l}{\epsilon}$$

Closed Formulas

- Defect Weyl anomaly governs defect contribution to the SUSY Casimir energy (dSCE) and Renyi entropy (dSRE)

$$E_{\mathcal{R}} = -\frac{1}{\omega_1}\left[\frac{d_2 - b}{6}\omega_2\omega_3 + \frac{2b - d_2}{24}\sigma_1\sigma_2\right]$$

$$S_n = \frac{2b - d_2}{12}\left[r_1 r_2\left(\frac{1}{n} - 1\right) + 2\right]\log\frac{l}{\epsilon}$$

$\omega_{1,2,3}$	chemical potentials conjugate to the 6d angular momentum
$\sigma_{1,2}$	chemical potentials conjugate to the 6d R-symmetry Cartans
$r_{1,2}$	weights of the R-symmetry chemical potentials

$$Z = \mathrm{Tr}\left[e^{-\beta\sum_{i=1}^{3}\omega_i J_i - \beta\sum_{a=1}^{2}\sigma_a R_a}\right]$$

Surface Defect Energy & Entropy

- Defect contribution to SUSY Casimir energy (dSCE)

$$\log \langle D \rangle = -\beta E_{\mathcal{R}} + O(\beta^0), \qquad \beta \to \infty$$

Low-temperature limit of partition function with defect insertion

- Defect contribution to the SUSY Rényi entropy (dSRE)

$$S_n = \frac{1}{1-n} \log \frac{Z_n}{(Z_1)^n}$$

Z_n: SUSY partition function on the n-replica space with defect insertion

Surface Defects in 6d (2,0) SCFTs

We study ½-BPS surface defect in 6d A_{N-1} (2,0) SCFT

Surface defect Weyl anomaly are labelled by representation

$$\langle t^{\mu}_{\mu} \rangle = -\frac{1}{24\pi}\left[\, b\, R^{\Sigma} + d_1 \Pi^{\mu}_{ab}\Pi^{ab}_{\mu} - d_2 W^{ab}_{ab}\right]$$

$$b = 24(\lambda,\rho) + 3(\lambda,\lambda) \qquad\qquad d_2 = 24(\lambda,\rho) + 6(\lambda,\lambda)$$

[Estes-Krym-O'Bannon-Robinson-Rodgers, 2018]

[Gentle-Gutperle-Marasinou, 2015]

Surface Defects in 6d (2,0) SCFTs

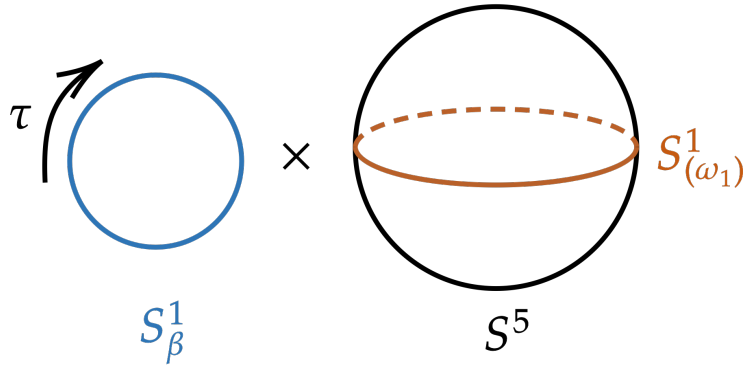
- No Lagrangian description.
- Non-perturbative method combining:

✓ **Localization**

✓ **Holography**

✓ **Anomaly polynomial**

Localization



Radial quantization:

Surface defect wrapping $S^1_\beta \times S^1_{(\omega_1)} \subset S^1_\beta \times S^5$

Dimensional reduction on S^1_β $\Longrightarrow$ 5d MSYM on S^5 with Wilson loop on $S^1_{(\omega_1)} \subset S^5$

$$\langle D \rangle = \frac{1}{Z}\int \mathrm{d}v \exp\left[\frac{2\pi}{\omega_1\omega_2\omega_3}\left(-\frac{\pi}{\beta}\sum_{i=1}^{N} v_i^2 + \frac{\sigma_1\sigma_2}{2}\sum_{j<i}(v_i - v_j)\right)\right]\mathrm{Tr}_{\mathcal{R}} \exp\left[\frac{2\pi v}{\omega_1}\right]$$

Saddle point approximation

$$E_{(k)} = -\frac{1}{2\omega_1}\left[\, k^2\, \omega_2\omega_3 + k(N-1)\, \sigma_1\sigma_2\right]$$

$$E_{[k]} = -\frac{1}{2\omega_1}\left[\, k\, \omega_2\omega_3 + k(N-k)\, \sigma_1\sigma_2\right]$$

$$E_{\mathcal{R}} = -\frac{1}{\omega_1}\left[\frac{d_2 - b}{6}\omega_2\omega_3 + \frac{2b - d_2}{24}\sigma_1\sigma_2\right]$$

Localization

dSCE on

$S^1_\beta \times S^1_n \subset S^1_\beta \times S^5_n$

governs $n \to 0$ behavior of

(via geometric relation)

dSRE on

$\mathbb{H}^1 \times S^1_n \subset \mathbb{H}^5 \times S^1_n$

$$E_{\mathcal{R}} = -\frac{1}{\omega_1}\left[\frac{d_2 - b}{6}\omega_2\omega_3 + \frac{2b - d_2}{24}\sigma_1\sigma_2\right]$$

$$\Rightarrow \quad S_{n\to 0} = -\frac{\mathrm{Vol}[\mathbb{H}^1]}{\mathrm{Vol}[S^1_\beta]}\beta E_{\mathcal{R}}\Big|_{n\to 0} = \frac{2b - d_2}{12}\frac{r_1 r_2}{n}\log\frac{l}{\epsilon}$$

$\omega_1 = 1/n$

$\omega_2 = \omega_3 = 1$

$\sigma_I\Big|_{n\to 0} = \frac{r_I}{n}$

Holography

Surface defect in fundamental representation $\quad \mathrm{AdS}_7/\mathrm{CFT}_6 \longleftrightarrow \quad$ Probe M2-brane in 7d 2-charged topological black hole

$$\log\langle D\rangle = -I_{M2} = \pi n T_2 r_H^2 \,\mathrm{Vol}[\mathbb{H}^1]$$

$\frac{2b-d_2}{12} = (N-1) \sim N$ at large N

$$S_n(N\to\infty) = \frac{\log\langle D\rangle - n\log\langle D\rangle|_{n=1}}{1-n} = N\left[r_1 r_2\left(\frac{1}{n}-1\right)+2\right]\log\frac{l}{\epsilon}$$

*Localization:

$$S_{n\to 0} = \frac{2b-d_2}{12}\,\frac{r_1 r_2}{n}\log\frac{l}{\epsilon}$$

$$\boxed{S_n = \frac{2b-d_2}{12}\left[r_1 r_2\left(\frac{1}{n}-1\right)+2\right]\log\frac{l}{\epsilon}}$$

Anomaly polynomial

- Depends on symmetry structures
- Encodes the information of anomaly

$$A_4 = -\frac{k_g}{24} p_1(T) + \frac{k_I}{2} c_2(F_I) + \frac{k_{IJ}}{2} c_1(F_I) c_1(F_J)$$

Gravitational anomaly ‘t Hooft anomaly Mixed anomaly

Anomaly polynomial

Even d-dim SCFT $\quad E = -\int_{\mathbb{R}^d} A_{d+2}$ [Bobev–Bullimore–Kim, 2015]

Generalize to surface defect $\quad E_{\mathcal{R}} = -\int_{\mathbb{R}^2} A_4$

Surface defect anomaly polynomial

$$A_4 = \frac{1}{2}(\lambda,\lambda)\big(c_2(L) - c_2(R)\big) + (\lambda,\rho)\big(c_2(I) - c_2(F)\big)$$ [Shimizu–Tachikawa, 2016]

$$\Longrightarrow \quad \boxed{E_{\mathcal{R}} = -\frac{1}{\omega_1}\left[\frac{d_2 - b}{6}\omega_2\omega_3 + \frac{2b - d_2}{24}\sigma_1\sigma_2\right]}$$

Anomaly polynomial

Extend with an additional U(1) symmetry $\tilde{E} = -\int_{\mathbb{R}^d \times \mathbb{R}^2} \tilde{A}_{d+2}$

SUSY Rényi entropy is given by the "extended Casimir energy"

$$S_n = \frac{n\tilde{E}|_{n=1} - \tilde{E}|_n}{1-n} \mathrm{Vol}[\mathbb{H}^1]$$

[Yankielowicz–Zhou, 2017]

Generalize to surface defect $\tilde{E}_{\mathcal{R}} = -\int_{\mathbb{R}^2 \times \mathbb{R}^2} \tilde{A}_4$

$$\Longrightarrow \quad \boxed{S_n = \frac{2b - d_2}{12}\left[r_1 r_2\left(\frac{1}{n} - 1\right) + 2\right]\log\frac{l}{\epsilon}}$$

Conclusion

- **Closed formulae** for surface defect contribution to SUSY Casimir energy and Rényi entropy and in 6d maximal supersymmetric CFT

$$E_{\mathcal{R}} = -\frac{1}{\omega_1}\left[\frac{d_2 - b}{6}\omega_2\omega_3 + \frac{2b - d_2}{24}\sigma_1\sigma_2\right]$$

➢ Controlled by surface defect Weyl anomaly

$$S_n = \frac{2b - d_2}{12}\left[r_1 r_2\left(\frac{1}{n} - 1\right) + 2\right]\log\frac{l}{\epsilon}$$

➢ Similar structure as 2d CFT

- **Non-perturbative methods** combining localization, holography and anomaly polynomial

Outlook

- Generalize to defects of different co-dimensions, bulk spacetime dimensions and symmetries:

  in 6d (1,0) SCFTs; Gukov-Witten defects in 4d SCFTs. [WIP]

- Defect Cardy formula? [WIP]

- More than one defect; defect intersections?